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CART-351

Project 1

Project Description: Interactive Data Exploration

For this project, my focus is to extract the data using the API of the weather around the world and Colorama to explore how the output of the data will look like with a visual and interactive approach.

The output of the project will consist of the user interacting with the data. The user (you) is searching for different conditions about the weather, specifically about the temperature and the air pollution levels. It will consist of the user needing to interact with the terminal, because it will ask their name and one for the city. The code will take the output of the data and will give a response about whether the city was found or not. If the city was not found, the user will need to try again until it matches both conditions and finds the perfect city to live in.The game doesn't have a certain limit to finding a city; the user can keep trying as many times they want, but they will also be able to quit the game at any time they want to.

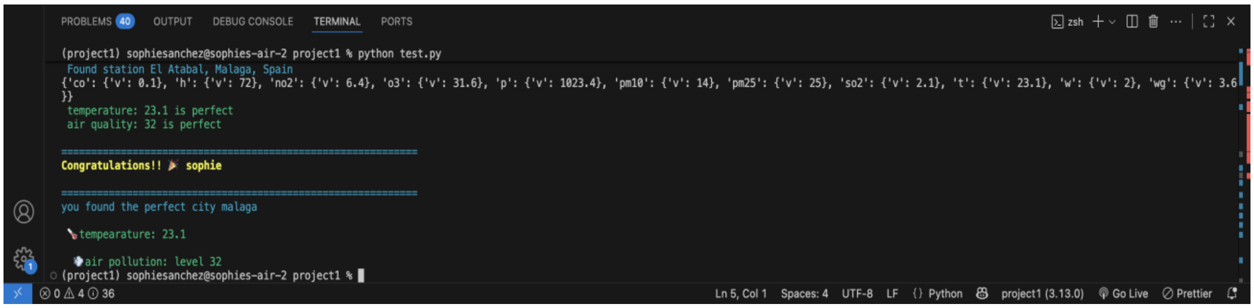
The intention and inspiration for this project were mostly to learn how to get information from the weather API in a more fun and interactive form, as well as being inspired by a family member who experienced different health issues because of the poor weather and air quality back in their country. When he moved here, the better weather conditions significantly improved his health. This experience showed me how important it is to find a place with good environmental conditions for one's wellbeing. This game gives the user a chance to search for and discover places that could be better for their health, just like my family member did.

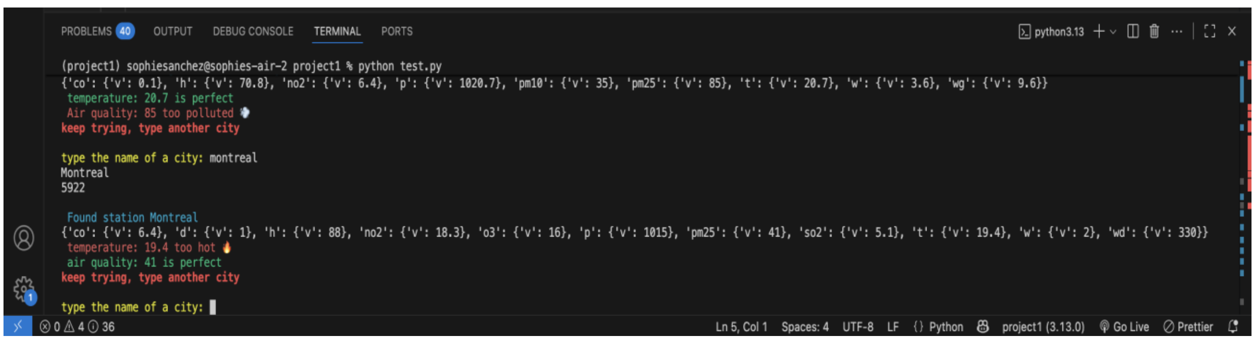
Also, I wanted to create something easier to understand and visually appealing. Since I specified the important information that I wanted to get from the API, this helped the user visualization of the data in a simpler approach. Also, I wanted the user to see the data information as a game. I think it's a good approach to make the user feel an emotional journey through trial and error, making the user input many different cities until they get the correct answer/input.

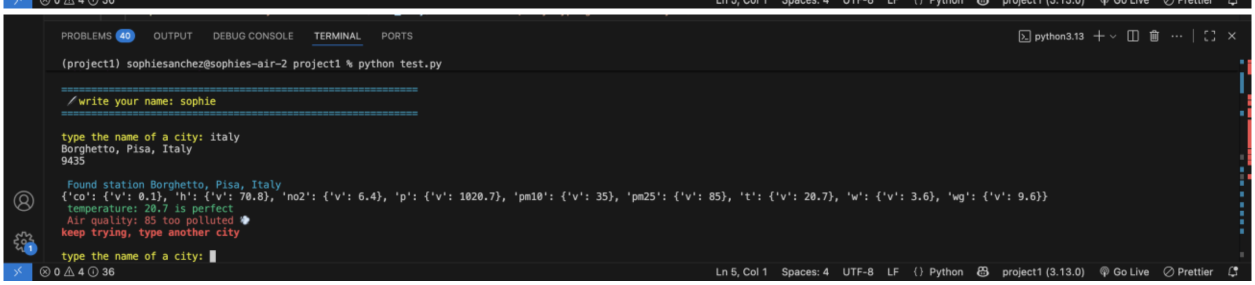
For this project, I worked with different resources: Colorama, emojis, and real-time pollution data from around the world. Since I wanted to make the information more readable for the user, this is why I decided to implement different colors depending on what information it was displaying. The use of colors for the pollution and temperature levels makes the game more visually appealing and helps communicate the data more effectively.

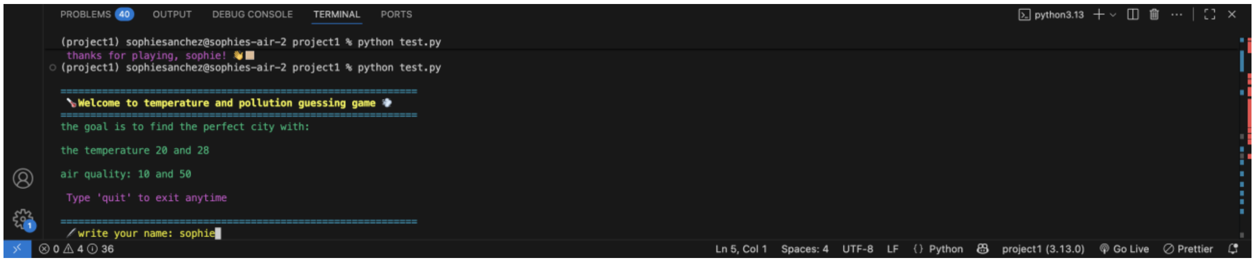
Ultimately, this project transforms weather data into an interactive experience that combines learning with play. It reflects how environmental conditions directly impact lives and health, while making the process of discovering this information more engaging and memorable. Through this work, I hope to show that data doesn't have to be difficult to understand, it can be colorful and interactive.

Screen Grabs:








PROBLEMS 40
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
python3.13
(project1) sophiesanchez@sophies-air-2 project1 % python test.py
thanks for playing, sophie!
(project1) sophiesanchez@sophies-air-2 project1 % python test.py
Welcome to temperature and pollution guessing game
the goal is to find the perfect city with:
the temperature 20 and 28
air quality: 10 and 50
Type 'quit' to exit anytime
write your name: sophie
Ln 5, Col 1
Spaces: 4
UTF-8
LF
Python
project1 (3.13.0)
Go Live
Prettier


test.py
#interaction: the user needs to put their name
print("\n" + Fore.CYAN + "="* 60)
name = input (Fore.YELLOW + " write your name: ")
print(Fore.CYAN + "="* 60)
#user_city = input("type the name of a city: ")
#looping
game_win= False
#looping starting game
while game_win == False:
user_city = input(Fore.YELLOW + "\ntype the name of a city: " + Style.RESET_ALL)
# if the user wants to leave the game, (this was made with the help of ressource)
if user_city.lower() == 'quit':
print(Fore.MAGENTA + f"\n thanks for playing, {name}! ")
break
#url for the wesite weatherApi to search te city
url = f"https://api.waqi.info/search/?token={token}&keyword={user_city}"
response= requests.get(url)
data= response.json()
#check the data available for the city, sometimes city is not available
if data['status'] != 'ok' or len(data['data']) == 0:
print(Fore.RED + f"your search for {user_city} was not found, try typing another city! ")
PROBLEMS 40
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
(project1) sophiesanchez@sophies-air-2 project1 % python test.py
type the name of a city: dubai
Dubai Shk. Mohd. Bin Zayed Road, UAE
Found station Dubai Shk. Mohd. Bin Zayed Road, UAE
temperature: 32 too hot
Air quality: 132 too polluted
keep trying, type another city
type the name of a city: quit
thanks for playing, sophie!
(project1) sophiesanchez@sophies-air-2 project1 %